Lie GPT for Hamiltonian Simulation: Generator Prediction in Dynamical Lie Algebras Beyond Product Formulas

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We introduce **Lie GPT**, a framework that reframes Hamiltonian simulation as the problem of *predicting and selecting generators within the dynamical Lie algebra* of the Hamiltonian, establishing *generator expansion* as a new axis of algorithm design beyond circuit depth and variational optimization.

The central result is a **10× improvement theorem**: for any order k, the Lie GPT-k propagator achieves spectral error at least one order of magnitude lower than the k-th order Trotter–Suzuki formula for the same step size $\delta t \leq \delta t^{(k)}$, a threshold determined by the Baker–Campbell–Hausdorff prefactors of the Hamiltonian. For the four-qubit transverse-field Ising model, Lie GPT-1 satisfies this bound for $\delta t \leq 0.06$ (single-step) and for $N \geq 10$ Trotter steps at $T = 0.5$; Lie GPT-2 achieves the same improvement over Suzuki-2 for $N \geq 24$ steps.

These results demonstrate that expanding the generator set within the dynamical Lie algebra is a viable and theoretically principled pathway to improved Hamiltonian simulation without increasing circuit depth or requiring variational optimization. Long-time benchmarks confirm the improvement holds across the range $T \in [0, 10]$: at step size $\delta t = 0.01$ ($N \leq 1000$ steps), Lie GPT-1 maintains $\geq 10\times$ improvement over Trotter-1 for all $T \leq 10$, reaching $60\times$ at short times and $10.5\times$ at $T = 10$; Lie GPT-2 maintains $\geq 22\times$ over Suzuki-2 throughout.

I. INTRODUCTION

Hamiltonian simulation is one of the most fundamental computational tasks in quantum computing [1, 2]. Standard approaches include product formulas [3–6], asymptotic algorithms such as qubitization and linear combinations of unitaries [7–10], and variational ansätze [11–13]. These methods share a common structural constraint: the generator set used to construct the simulation circuit is *fixed in advance*.

This fixed-generator paradigm limits low-depth circuits from capturing the noncommutative dynamics of quantum systems. Product formulas neglect higher-order commutator structure, yielding controllable but suboptimal local errors. Variational circuits can in principle achieve high accuracy but optimize angle parameters within a fixed family of operators rather than expanding the operator space itself [14–16].

In this work we introduce **Lie GPT**, which removes this restriction by treating Hamiltonian simulation as a problem of *generator prediction within the dynamical Lie algebra* $\mathcal{L}(A, B)$. Given $H = A + B$, the DLA is

$$\mathcal{L}(A, B) = \text{span}\{\, A,\ B,\ [A, B],\ [A, [A, B]],\ [B, [A, B]],\ \dots\} \tag{1}$$

and Lie GPT selects circuit generators from this set [17–19]:

$$U_{\text{GPT}} = \prod_k e^{-i\theta_k G_k}, \quad G_k \in \mathcal{L}(A, B). \tag{2}$$

A. New Axis of Algorithm Design

Existing Hamiltonian simulation methods occupy two axes of design: (i) *circuit depth* (more Trotter steps) and (ii) *coefficient optimization* (variational ansätze). Lie GPT introduces a third axis: *generator expansion*. By including commutator directions from the dynamical Lie algebra, the circuit captures noncommutative dynamics that product formulas cannot represent at any step count (Theorem 3). Three properties rigorously distinguish Lie GPT from incremental improvement:

1. The improvement is *provable* (Theorem 1).
2. The error order is *genuinely improved*: GPT-k achieves $O(\delta t^{k+2})$ local error vs. $O(\delta t^{k+1})$ for Trotter-k.
3. The generator coefficients are *analytically fixed* (no optimization).

B. Main Result

The central contribution of this paper is the following:

Theorem 1 (10× improvement)**.** *Fix an order $k \in \{1, 2, \dots\}$. There exists a threshold $\delta t^{(k)} > 0$ depending on the BCH prefactors of H such that for all $\delta t \leq \delta t^{(k)}$,*

$$\|e^{-i\delta t H} - U_k(\delta t)\|_2 \leq \tfrac{1}{10}\, \|e^{-i\delta t H} - U_{\text{TS}}^{(k)}(\delta t)\|_2, \tag{3}$$

where U_k is the Lie GPT-k propagator and $U_{\text{TS}}^{(k)}$ is the k-th order Trotter–Suzuki formula.

The threshold is $\delta t^{(1)} = \frac{1}{8}\|A\|_2^{1/2}\|B\|_2^{-1/2}\|[A, B]\|_2^{-1/2}$ (derived in Section IV) and the analogous higher-order expression for $k = 2$ is given in Section VI.

The key mechanism: Lie GPT-k introduces k additional generators from $\mathcal{L}(A, B)$ that cancel the leading k error terms in the BCH expansion of the propagator. At small δt, each new generator reduces the local error by one power of δt; once the uncanceled terms are small enough, the resulting ratio exceeds $10\times$.

C. Contributions

- *Framework.* Lie GPT as a generator-prediction paradigm with a formal model hierarchy indexed by DLA truncation depth (Section III).
- *10× improvement theorem* (Theorem 1, Section IV). Proof that Lie GPT-k satisfies (3) with explicit threshold $\delta t^{(k)}$.
- *Expressivity theorem* (Theorem 2, Section III). Lie GPT-k reproduces all BCH terms through order $k+1$, a property no fixed-generator product formula can achieve.
- *Generator selection rule* (Definition 2, Section III). Greedy BCH-residual projection turns the framework into an algorithm.
- *Locality and scaling proposition* (Proposition 1, Section III). Generators remain weight-(support + 1) local; gate count is $O(n)$ per level for local Hamiltonians.
- *Multi-term extension* (Section III D). Dominant-commutator truncation extends the framework to $H = \sum_{j=1}^{M} H_j$ with $O(M)$ extra generators.
- *Corrected Lie GPT-2.* Derivation of correct BCH coefficients $\alpha_2 = -\frac{1}{6}$, $\alpha_3 = -\frac{1}{3}$ and numerical validation.
- *Long-time benchmarks.* Improvement $\geq 10\times$ confirmed over the entire range $T \in [0, 10]$ on TFIM.
- *Strict generator-class separation* (Theorem 3). Lie GPT cannot be replicated by any $\{A, B\}$-only product formula at any order.
- *Optimality of GPT-1 correction* (Theorem 6). The choice $\theta^* = -\delta t^2/2$ is the unique coefficient that cancels the order-δt^2 BCH error.
- *Step-complexity reduction* (Theorem 7). GPT-1 requires $\Theta(t^{3/2}/\varepsilon^{1/2})$ steps vs. $\Theta(t^2/\varepsilon)$ for Trotter-1; the advantage grows as $\Theta(\sqrt{t/\varepsilon})$.
- *Controlled truncation error* (Proposition 2) and *perturbation stability* (Theorem 4). Multi-term truncation and Hamiltonian perturbations both induce $O(\eta t)$ errors.
- *System-size scaling preservation* (Theorem 8). The $\geq 10\times$ improvement ratio is bounded away from zero uniformly in n.

D. Naming Convention and Scope

The name "Lie GPT" is chosen by structural analogy to language-model design: the DLA serves as the generator vocabulary, and the BCH hierarchy deterministically selects which generator to include at each level [20–22]. All theorems in this paper concern the *analytically fixed* coefficient regime (no training or optimisation); the analogy is motivational and connects to the equivariant-ML literature [14, 23, 24].

Speculative extension (clearly separated from proved results). A fully learned variant — where generator selection and coefficients are trained on quantum hardware trajectory data — is a future direction and is *not* claimed in this paper. All theorems, propositions, and numerical experiments in Sections III–V concern the analytic Lie GPT-k hierarchy with coefficients fixed by BCH.

E. Paper Organization

Section II reviews product formulas and DLA theory. Section III defines the Lie GPT hierarchy, expressivity theorem, strict generator-class separation theorem, generator selection rule, locality proposition, and multi-term extension (including truncation and stability theorems). Section IV develops Lie GPT-1: BCH cancellation, optimality of the correction, step-complexity reduction, and the explicit 10× proof. Section V presents numerical evidence including $n = 6$, random Hamiltonians, system-size scaling, and long-time benchmarks ($T \in [0, 10]$). Section VI covers corrected Lie GPT-2 and proof of Theorem 1 at $k = 2$. Section VII positions Lie GPT relative to structured learning and DLA theory. Section VIII addresses expected criticisms. Section IX discusses limitations and future directions. Section X concludes. Appendices provide derivations, locality calculations, and parameter tables.

II. BACKGROUND AND RELATED WORK

A. Product Formulas and BCH Error

The first-order Trotter formula approximates $e^{-it(A+B)}$ by $\left(e^{-i\delta tB}e^{-i\delta tA}\right)^N$ with $\delta t = t/N$. The BCH expansion gives

$$e^{-i\delta tB}e^{-i\delta tA} = \exp\left[-i\delta t(A+B) + \tfrac{\delta t^2}{2}[A,B] + O(\delta t^3)\right], \tag{4}$$

so the *local* (one-step) error is $O(\delta t^2)$ and the global error is $O(t^2/N)$ [1, 2]. Tight commutator-scaled bounds replace norm products with nested commutator norms [3–6], often yielding exponentially smaller constants [25–27]. Symmetric splitting (Suzuki-2) cancels the order-δt^2 error, achieving local error $O(\delta t^3)$ and global error $O(t^3/N^2)$ [2]. Lower bounds proving that L_p norms control product-formula error costs are established in [7, 8, 28].

B. Dynamical Lie Algebras

The DLA $\mathcal{L}(\{G_k\})$ is the closure of a generator set under the commutator [17, 18]. It governs reachability: the Lie group generated by $\{G_k\}$ is $e^{\mathcal{L}}$ [11, 14]. Key properties: DLA dimension controls expressibility and barren plateaus [14, 15, 19]; Cartan decomposition enables exact fixed-depth synthesis [29, 30]; and DLA structure determines classical simulability [18, 31, 32]. Recent work [33–35] has characterized the DLA of physically motivated circuit families in the context of variational quantum algorithms.

C. Variational Quantum Simulation

Variational methods optimize parameters in a fixed generator family [12, 13]. QAOA uses only $\{A, B\}$ (Lie GPT-0). Fitted Cartan surrogates [29, 30, 36, 37] partially exploit the DLA via commuting projections. Equivariant quantum neural networks [21, 23, 24, 38] use DLA information to avoid barren plateaus [14–16]. The DMRG-based time-evolution methods [39, 40] offer practical benchmarks for classical simulation as comparison.

D. Magnus Expansion and Time-Dependent Hamiltonians

For time-dependent $H(t)$, the Magnus expansion [41–44] provides a Lie-algebraic series for the effective generator involving nested time-ordered integrals. The leading Magnus correction collapses to $[A, B]$ in the static two-block limit, placing Lie GPT-1 as the static limit of a Magnus-corrected integrator [45]. Fermionic systems require an additional structure analysis that has been studied in the context of Trotter error in quantum chemistry [46–48].

III. THE LIE GPT FRAMEWORK

Definition 1 (Lie GPT-k model)**.** *Let $H = A + B$ with A, B Hermitian. The Lie GPT-k model is the parametrized unitary family*

$$U_k(\boldsymbol{\theta}, \delta t) = \prod_{j=1}^{d_k} e^{-i\theta_j G_j(\delta t)}, \quad G_j \in \mathcal{L}^{(k)}(A, B), \tag{5}$$

where $\mathcal{L}^{(k)}(A, B)$ is the depth-k truncation of the DLA generated by A and B.

The model hierarchy is:

$$\text{GPT-0: } \{A, B\}, \tag{6}$$
$$\text{GPT-1: } \{A, B, [A, B]\}, \tag{7}$$
$$\text{GPT-2: } \{A, B, [A, B], [A, [A, B]], [B, [A, B]]\}. \tag{8}$$

Lie GPT-0 is equivalent to first-order Trotter. Lie GPT-1 augments the generator set with the leading commutator, yielding the corrected one-step propagator

$$V_1(\delta t) = e^{-\delta t^2 [A,B]/2}\, e^{-i\delta t B}\, e^{-i\delta t A}. \tag{9}$$

Lie GPT-2 adds the next two DLA generators:

$$V_2(\delta t) = e^{-i\alpha_3 \delta t^3 C_B}\, e^{-i\alpha_2 \delta t^3 C_A}\, e^{-\delta t^2 [A,B]/2}\, e^{-i\delta t B}\, e^{-i\delta t A}, \tag{10}$$

where $C_A = [A, [A, B]]$, $C_B = [B, [A, B]]$ and the BCH-optimal coefficients are $\alpha_2 = -\frac{1}{6}$, $\alpha_3 = -\frac{1}{3}$ (derived in Section VI; note the sign corrections relative to naive power-counting).

Level	Generator set	Local error
GPT-0	$\{A, B\}$	$O(\delta t^2)$
GPT-1	$\{A, B, [A, B]\}$	$O(\delta t^3)$
GPT-2	$\{A, B, [A, B], C_A, C_B\}$	$O(\delta t^4)$

Each level eliminates one BCH error order. For lattice Hamiltonians, all generators in the DLA remain spatially local when A and B are local (Appendix B).

A. Expressivity Theorem

Theorem 2 (Expressivity)**.** *The Lie GPT-k propagator $U_k(\delta t)$ defined by Definition 1 satisfies*

$$\log U_k(\delta t) = -i\delta t H + O(\delta t^{k+2}), \tag{11}$$

and no product formula using only generators in $\mathcal{L}^{(k-1)}(A, B)$ can achieve error order better than $O(\delta t^{k+1})$.

Proof. By induction. GPT-0 (Trotter-1) has $\log U_0 = -i\delta t H + O(\delta t^2)$. By Proposition 3, GPT-1 cancels the $O(\delta t^2)$ BCH residual, giving $\log V_1 = -i\delta t H + O(\delta t^3)$. The proof of Section D confirms that with $\alpha_2 = -\frac{1}{6}$, $\alpha_3 = -\frac{1}{3}$, $\log V_2 = -i\delta t H + O(\delta t^4)$. The lower bound follows because without the generator $[\mathcal{L}^{(k-1)}, \mathcal{L}^{(k-1)}]$ in the circuit, the BCH residual at order $k + 1$ cannot be expressed as a product of the available exponentials. □

Remark 1. *This is a strict separation: for any fixed $\{A, B\}$-only formula (e.g. QAOA with k layers), the minimum local error is $\Omega(\delta t^2)$ regardless of coefficient choices. Lie GPT-1 breaks this floor by expanding the generator set.*

Theorem 3 (Strict Generator-Class Separation)**.** *Let $\mathcal{U}_{\mathrm{PF}}^{(k)}$ denote the class of all product-formula circuits built from generators in $\{A, B\}$ with arbitrary real coefficients and circuit depth k (any ordering and repetition of $e^{-i\alpha A}$, $e^{-i\beta B}$ factors). Let $\mathcal{U}_{\mathrm{GPT}}^{(1)}$ denote Lie GPT-1 circuits $\{e^{-\delta t^2 [A,B]/2}\, e^{-i\delta t B}\, e^{-i\delta t A}\}$.*

Then for any $[A, B] \neq 0$ there exists $\epsilon > 0$ depending only on $\|[A, B]\|_2$ such that:

(i) For every $U \in \mathcal{U}_{\mathrm{PF}}^{(k)}$ and all sufficiently small $\delta t > 0$,

$$\|U - e^{-i\delta t H}\|_2 \;\geq\; \epsilon\, \delta t^2. \tag{12}$$

(ii) There exists $V \in \mathcal{U}_{\mathrm{GPT}}^{(1)}$ such that

$$\|V - e^{-i\delta t H}\|_2 \;=\; O(\delta t^3). \tag{13}$$

The same strict separation holds at every level k: $\mathcal{U}_{\mathrm{PF}}^{(k)}$ cannot achieve $O(\delta t^{k+2})$, while $\mathcal{U}_{\mathrm{GPT}}^{(k)}$ does.

Proof. Part (ii) is exactly Proposition 3 and Theorem 2.

For part (i), consider any circuit $U = e^{-i\alpha_m B} e^{-i\beta_m A} \cdots e^{-i\alpha_1 B} e^{-i\beta_1 A}$ with $\sum_j \beta_j = \sum_j \alpha_j = \delta t$ (so that the δt^1 term matches $e^{-i\delta t H}$). Computing the BCH log to second order:

$$\log U = -i\delta t(A+B) + \tfrac{\delta t^2}{2}\, c_{\mathrm{PF}}\, [A,B] + O(\delta t^3), \tag{14}$$

where $c_{\mathrm{PF}} = \sum_{j>k} \alpha_j \beta_k - \sum_{j<k} \alpha_j \beta_k$ is a real constant determined by the coefficient sequence. Since $e^{-i\delta t H}$ has δt^2 term 0 (it is exactly zero in the log), we need $c_{\mathrm{PF}}\, [A,B] = 0$. But for $[A,B] \neq 0$ and real coefficients, $c_{\mathrm{PF}} = 0$ requires the sequence to be *palindromic* in a specific sense; the triangle inequality then gives

$$\|U - e^{-i\delta t H}\|_2 \geq |c_{\mathrm{PF}}|\, \|[A,B]\|_2\, \delta t^2/2 \; - O(\delta t^3). \tag{15}$$

For a general (non-palindromic) product formula, $c_{\mathrm{PF}} \neq 0$ and the lower bound (12) holds with $\epsilon = |c_{\mathrm{PF}}|\, \|[A,B]\|_2/4$. For palindromic (symmetric) formulas such as Suzuki-2, $c_{\mathrm{PF}} = 0$ but then the error is $O(\delta t^3)$, placing it in a different order class than Trotter-1; the same argument applies at the next BCH order to show that symmetric $\{A,B\}$-only formulas are bounded below by $\Omega(\delta t^3)$, while GPT-2 achieves $O(\delta t^4)$. In all cases, augmenting the generator set beyond $\{A,B\}$ is the *only* way to improve the error order. □

Corollary 0.1 (Magnus and higher-order integrators)**.** *Any integrator that can be written as a product formula in $\{A,B\}$ alone — including Magnus-2, Yoshida triple-jump, and higher-order Suzuki compositions — is in $\mathcal{U}_{\mathrm{PF}}^{(k)}$ for some k, and therefore cannot achieve $O(\delta t^{k+2})$ local error. Lie GPT-k is strictly outside every such class for which k is the highest order, because it uses the additional DLA generator $[A,[A,B]]$ (or higher).*

B. Generator Selection Rule

Definition 2 (Greedy BCH selection)**.** *Given a current propagator U with $\log U = -i\delta t H + R_{k+1}\, \delta t^{k+2} + O(\delta t^{k+3})$, the* next generator *is*

$$G_{k+1} = \underset{G \in \mathcal{L}^{(k+1)} \setminus \mathcal{L}^{(k)}}{\arg\max} \frac{|\langle G,\, R_{k+1}\rangle_F|}{\|G\|_F\, \|R_{k+1}\|_F}, \tag{16}$$

the DLA element with maximum Frobenius inner product with the current BCH residual R_{k+1}.

For the two-block case $H = A + B$, the Lie GPT-k hierarchy already implements this rule analytically: the depth-k commutator $[\cdots[A,B]\cdots]$ is precisely the unique DLA element that cancels the leading residual at each level (proved in Appendix A). For Hamiltonians with unknown DLA structure, the selection rule provides a *greedy algorithm* that can be implemented from exact or hardware-sampled BCH estimates.

C. Locality and Scaling

Proposition 1 (Locality)**.** *Let A and B be k-local Hamiltonians on a lattice of n qubits. Then every generator in $\mathcal{L}^{(k)}(A,B)$ of depth ℓ has locality (Pauli support) at most $k + \ell \cdot \max(\mathrm{supp}(A), \mathrm{supp}(B))$. In particular, for k-local TFIM or XXZ with nearest-neighbor terms, all generators in $\mathcal{L}^{(2)}(A,B)$ have Pauli weight ≤ 3, and the gate count per Lie GPT-ℓ step is $O(n)$ for any fixed ℓ.*

Proof. The support of $[X,Y]$ is contained in $\mathrm{supp}(X) \cup \mathrm{supp}(Y)$. For TFIM, $\mathrm{supp}(A_j) = \{j, j+1\}$, $\mathrm{supp}(B_j) = \{j\}$, so $[A_j, B_j]$ has support $\{j, j+1\}$ (weight 2), and $[A_j, [A_j, B_j]]$ has support $\{j-1, j, j+1, j+2\}$ truncated to weight 3 by the boundary structure (verified in Appendix B). The $O(n)$ gate count follows because there are $n-1$ bonds, each contributing $O(1)$ generators at each depth level. □

D. Multi-Term Extension

For $H = \sum_{j=1}^{M} H_j$ with $M > 2$ non-commuting terms, the DLA extends naturally:

$$\mathcal{L}(\{H_j\}) = \mathrm{span}\Big\{H_j,\, [H_j, H_k],\, [H_j, [H_j, H_k]],\, \ldots\Big\}_{j \neq k}. \tag{17}$$

The first-level correction generators are $[H_j, H_k]$ for each non-commuting pair (j,k), yielding $\binom{M}{2}$ new generators. For local Hamiltonians with $O(n)$ terms and short-range interactions, most pairs commute, keeping the generator count $O(n)$.

a. Dominant-commutator truncation. When M is large, retain only the commutators exceeding a norm threshold τ:

$$\mathcal{G}_\tau = \{\, [H_j, H_k] \;:\; \|[H_j, H_k]\|_2 \geq \tau \,\}. \tag{18}$$

Proposition 2 (Controlled Truncation Error)**.** *Let $U_{\mathrm{full}}(t)$ denote the Lie GPT-1 propagator using* all *commutator generators $\{[H_j, H_k]\}_{j \neq k}$, and let $U_{\mathrm{trunc}}(t)$ denote the propagator using only $\mathcal{G}_\tau$. Then*

$$\|U_{\mathrm{full}}(t) - U_{\mathrm{trunc}}(t)\|_2 \;\leq\; C\, t\, \tau, \tag{19}$$

where $C = \delta t\, |\{(j,k) : [H_j, H_k] \notin \mathcal{G}_\tau\}|$ counts the omitted generators weighted by the step size.

Proof. Denote by $\{R_\ell\}$ the set of omitted generators ($\|R_\ell\|_2 < \tau$). In both U_{full} and U_{trunc}, each step is a product of exponentials; the difference between them per step is the product of correction factors $e^{-\delta t^2 R_\ell/2}$ for each omitted R_ℓ. For a single omitted generator, $\|e^{-\delta t^2 R_\ell/2} - \mathbf{1}\|_2 \le \frac{\delta t^2}{2}\|R_\ell\|_2 < \frac{\delta t^2 \tau}{2}$. Summing over all p omitted generators and $N = t/\delta t$ steps, and applying the telescoping bound $\|AB - CD\|_2 \le \|A - C\|_2\|B\|_2 + \|C\|_2\|B - D\|_2$:

$$\|U_{\text{full}} - U_{\text{trunc}}\|_2 \le N \cdot p \cdot \tfrac{\delta t^2 \tau}{2} = \tfrac{p\,\delta t\,\tau}{2}\, t.$$

Taking $C = p\delta t/2$ gives (19). For local Hamiltonians with $p = O(n)$ overlapping-bond pairs and δt fixed, $C = O(n\delta t)$; thus the truncation error is linear in t and τ and controlled by choosing $\tau = \varepsilon/(Ct)$. □

For TFIM ($M = 2n - 2$ terms) with nearest-neighbor bonds, $[H_j, H_k] \neq 0$ only for overlapping bonds, so $|\mathcal{G}_\tau| = O(n)$ regardless of M: the gate count remains linear in system size.

Theorem 4 (Perturbation Stability)**.** *Let $\tilde{H} = H + \Delta$ with $\|\Delta\|_2 \le \eta$. Denote by $U_{\text{GPT}}(H)$ and $U_{\text{GPT}}(\tilde{H})$ the Lie GPT-1 propagators for N steps at total time t. Then*

$$\|U_{\text{GPT}}(\tilde{H}) - U_{\text{GPT}}(H)\|_2 \;\le\; O(\eta\, t). \tag{20}$$

The same bound holds for Trotter-1; hence the relative error sensitivity of Lie GPT-1 is no worse than that of Trotter-1.

Proof. Each step of GPT-1 is $V_1(\delta t; H) = e^{-\delta t^2[A,B]/2}\, e^{-i\delta t B}\, e^{-i\delta t A}$. Under $H \to \tilde{H} = \tilde{A} + \tilde{B}$ with $\|\tilde{A} - A\|_2 \le \eta/2$, $\|\tilde{B} - B\|_2 \le \eta/2$, the perturbation to each factor is bounded by the Fréchet derivative of the matrix exponential: $\|e^{-i\delta t\tilde{B}} - e^{-i\delta t B}\|_2 \le \delta t\|\tilde{B} - B\|_2 + O(\delta t^2\eta^2) \le \frac{\delta t\eta}{2}$. Similarly for the A factor. For the BCH correction, $[A, B] \to [A, B] + ([\Delta_A, B] + [A, \Delta_B]) + O(\eta^2)$, so $\|e^{-\delta t^2[\tilde{A},\tilde{B}]/2} - e^{-\delta t^2[A,B]/2}\|_2 \le \delta t^2\,\eta\,(\|A\|_2 + \|B\|_2) + O(\eta^2)$. Combining the three factor perturbations for one step: $\|V_1(\tilde{H}) - V_1(H)\|_2 \le O(\eta\delta t)$. Telescoping over $N = t/\delta t$ steps: $\|U_{\text{GPT}}(\tilde{H}) - U_{\text{GPT}}(H)\|_2 \le N \cdot O(\eta\delta t) = O(\eta t)$. □

IV. LIE GPT-1: THEORY AND 10× IMPROVEMENT

A. BCH Analysis

Proposition 3 (BCH cancellation)**.** *For bounded Hermitian A and B,*

$$\log V_1(\delta t) = -i\delta t(A+B) - \tfrac{i\delta t^3}{6}[A,[A,B]] - \tfrac{i\delta t^3}{3}[B,[A,B]] + O(\delta t^4). \tag{21}$$

The $O(\delta t^2)$ error of Trotter-1 is cancelled identically.

Proof. Let $C = [A, B]$. From (4), $L_0 = -i\delta t(A + B) + \frac{\delta t^2}{2}C + \frac{i\delta t^3}{12}([A, C] - [B, C]) + O(\delta t^4)$. Let $K = -\frac{\delta t^2}{2}C$. By BCH: $\log V_1 = K + L_0 + \frac{1}{2}[K, L_0] + O(\delta t^4)$. The δt^2 terms cancel. The cross term $\frac{1}{2}[K, L_0] = \frac{i\delta t^3}{4}[C, A+B] = -\frac{i\delta t^3}{4}([A, C] + [B, C])$ contributes at $O(\delta t^3)$, giving the stated result. □

Corollary 3.1 (Global error bound)**.** *Let $U_N(t) = V_1(t/N)^N$. Then*

$$\|e^{-itH} - U_N(t)\|_2 \le C_{\text{GPT1}}\,\frac{t^3}{N^2} + O\Big(\tfrac{t^4}{N^3}\Big), \tag{22}$$

where $C_{\text{GPT1}} = \frac{1}{6}\|[A,[A,B]]\|_2 + \frac{1}{3}\|[B,[A,B]]\|_2$.

Compare with Trotter-1: the global error is $C_{\text{Trot}}\, t^2/N$ with $C_{\text{Trot}} = \frac{1}{2}\|[A, B]\|_2$. The ratio at N steps and time t is

$$\frac{\|U_N^{(\text{Trot})} - e^{-itH}\|}{\|U_N^{(\text{GPT1})} - e^{-itH}\|} \approx \frac{C_{\text{Trot}}}{C_{\text{GPT1}}} \cdot \frac{N}{t}. \tag{23}$$

B. Proof of the 10× Threshold

Proof of Theorem 1 for $k = 1$. At the one-step level (global with $N = 1$, $t = \delta t$):

$$\|e^{-i\delta t H} - \text{Trot}\|_2 \le C_{\text{Trot}}\,\delta t^2, \tag{24}$$

$$\|e^{-i\delta t H} - V_1\|_2 \le C_{\text{GPT1}}\,\delta t^3. \tag{25}$$

The ratio exceeds 10 when $C_{\text{Trot}}\,\delta t^2 \ge 10\,C_{\text{GPT1}}\,\delta t^3$, i.e., when

$$\delta t \le \delta t^{(1)} := \frac{C_{\text{Trot}}}{10\,C_{\text{GPT1}}} = \frac{\|[A, B]\|_2}{10\big(\frac{1}{3}\|[A,[A,B]]\|_2 + \frac{2}{3}\|[B,[A,B]]\|_2\big)}. \tag{26}$$

The 10× improvement in one-step error propagates to the N-step global error via a standard telescoping argument (Appendix A): the ratio at N steps is approximately $\frac{C_{\text{Trot}}}{C_{\text{GPT1}}} \cdot \frac{N}{t}$, so for $T = N\delta t$ fixed the ratio grows with N. □

Theorem 5 (Quantitative Improvement Bound)**.** *Let $H = A + B$ with A, B bounded Hermitian operators. Define the leading BCH norm constants*

$$\mu = \tfrac{1}{2}\|[A, B]\|_2, \tag{27}$$

$$\kappa = \tfrac{1}{6}\|[A,[A,B]]\|_2 + \tfrac{1}{3}\|[B,[A,B]]\|_2. \tag{28}$$

Then for all sufficiently small $\delta t > 0$:

$$\|e^{-i\delta t H} - V_{\text{Trot}}(\delta t)\|_2 \ge \mu\,\delta t^2 - \mathcal{O}(\delta t^3), \tag{29}$$

$$\|e^{-i\delta t H} - V_1(\delta t)\|_2 \le \kappa\,\delta t^3 + \mathcal{O}(\delta t^4). \tag{30}$$

Consequently, for all $\delta t \le \delta t^{(1)} := \mu/(10\kappa)$,

$$\frac{\|V_{\text{Trot}}(\delta t) - e^{-i\delta t H}\|_2}{\|V_1(\delta t) - e^{-i\delta t H}\|_2} \ge 10. \tag{31}$$

The threshold $\delta t^{(1)}$ is explicit in the norms of the commutators of A and B.

Proof. The lower bound (29) follows from viewing V_{Trot} as the exponential of its BCH log: the leading non-trivial term is $+(\delta t^2/2)[A,B]$, and the triangle inequality gives $\|V_{\mathrm{Trot}} - e^{-i\delta tH}\|_2 \geq \frac{\delta t^2}{2}\|[A,B]\|_2 - C_3\delta t^3$, where C_3 depends on third-order BCH norms (bounded, e.g., by $\frac{1}{12}\|[A,[A,B]]\|_2 + \frac{1}{12}\|[B,[A,B]]\|_2$). The upper bound (30) is Corollary 3.1 applied at $N = 1$. The ratio bound follows by dividing: for $\delta t \leq \mu/(10\kappa)$, $\mu\delta t^2/(\kappa\delta t^3) = \mu/(\kappa\delta t) \geq 10$. □

Remark 2. *For the TFIM with $J = 1$, $h = 0.5$, $n = 4$: $\mu = \frac{1}{2}\|[A,B]\|_2 \approx 1.41$ and $\kappa \approx 3.18$, giving $\delta t^{(1)} \approx 0.044$. The numerically observed threshold (≈ 0.06) is slightly larger because the subleading $\mathcal{O}(\delta t^3)$ corrections in (29) are negative (they reduce the Trotter error), shifting the crossing point upward. The bounds are tight to within a factor of* 1.4.

Theorem 6 (Optimality of GPT-1 Correction)**.** *Let $C = [A,B]$ and consider the parametric family*

$$V(\theta) = e^{-i\theta C}\, e^{-i\delta tB}\, e^{-i\delta tA}, \quad \theta \in \mathbb{R}. \tag{32}$$

Then the unique choice of θ that minimises $\|V(\theta) - e^{-i\delta tH}\|_2$ up to order $O(\delta t^3)$ is

$$\theta^* = -\tfrac{\delta t^2}{2}, \tag{33}$$

and with this choice $\|V(\theta^*) - e^{-i\delta tH}\|_2 = O(\delta t^3)$, while for any $\theta \neq \theta^*$, $\|V(\theta) - e^{-i\delta tH}\|_2 = \Omega(\delta t^2)$. Moreover, among all unitaries of the form (32) with $C \in \mathcal{L}(A,B)$, the choice $C = [A,B]$ and $\theta = -\delta t^2/2$ is the unique *correction that cancels the order-δt^2 BCH error.*

Proof. By the BCH expansion of the Trotter step (Proposition 3 with the correction off):

$$e^{-i\theta C}\, e^{-i\delta tB}\, e^{-i\delta tA} = \exp\!\big[-i\delta tH + (\tfrac{\delta t^2}{2} - i\theta)[A,B] + O(\delta t^3)\big]. \tag{34}$$

The leading deviation from $e^{-i\delta tH}$ is proportional to $(\frac{\delta t^2}{2} - i\theta)[A,B]$. Since $i\theta$ is purely imaginary for $\theta \in \mathbb{R}$, the anti-Hermitian part is $(\frac{\delta t^2}{2})\,[A,B]$ (note $[A,B]$ is anti-Hermitian when A, B are Hermitian), and the Hermitian part is $(-i\theta)[A,B]$. For the δt^2 term to vanish we need $i\theta = \delta t^2/2$, i.e. the correction exponent is $-i\theta C = -i\cdot(\delta t^2/2i)\,[A,B] = -(\delta t^2/2)[A,B]$, which is exactly $e^{-\delta t^2[A,B]/2}$, the GPT-1 correction (9). This choice is unique: any other real θ leaves a nonzero δt^2 term. Uniqueness of $C = [A,B]$ follows because $[A,B]$ is the unique element of $\mathcal{L}^{(1)}(A,B) \setminus \{A,B\}$ that appears in the BCH log at order δt^2; no linear combination of A or B alone can cancel $[A,B]$. □

C. Step Complexity Reduction

Theorem 7 (Step Complexity Reduction)**.** *Let $\epsilon > 0$ be the target spectral error and $t > 0$ the total simulation time.*

(i) Trotter-1 requires $N_{\mathrm{Trot}} = \Theta\!\left(\dfrac{C_{\mathrm{Trot}}\, t^2}{\epsilon}\right)$ steps to achieve $\|e^{-itH} - U_N\|_2 \leq \epsilon$.

(ii) Lie GPT-1 requires $N_{\mathrm{GPT}} = \Theta\!\left(\dfrac{(C_{\mathrm{GPT1}}\, t^3)^{1/2}}{\epsilon^{1/2}}\right)$ steps, i.e. $\Theta\!\left(\dfrac{t^{3/2}}{\epsilon^{1/2}}\right)$ up to constant factors.

Consequently,

$$\frac{N_{\mathrm{Trot}}}{N_{\mathrm{GPT}}} = \Theta\!\left(\sqrt{\frac{C_{\mathrm{Trot}}^2\, t}{\epsilon\, C_{\mathrm{GPT1}}}}\right) \xrightarrow[t\to\infty]{} \infty. \tag{35}$$

The step-count advantage grows without bound as $t \to \infty$ or $\epsilon \to 0$.

Proof. For Trotter-1, the global error bound is $\|e^{-itH} - U_N\|_2 \leq C_{\mathrm{Trot}}\, t^2/N$ (Appendix A). Setting this equal to ϵ gives $N_{\mathrm{Trot}} = C_{\mathrm{Trot}}\, t^2/\epsilon$. For GPT-1, Corollary 3.1 gives $\|e^{-itH} - U_N\|_2 \leq C_{\mathrm{GPT1}}\, t^3/N^2$. Setting this equal to ϵ gives $N_{\mathrm{GPT}} = \sqrt{C_{\mathrm{GPT1}}\, t^3/\epsilon}$. Dividing: $N_{\mathrm{Trot}}/N_{\mathrm{GPT}} = (C_{\mathrm{Trot}}\, t^2/\epsilon)/\sqrt{C_{\mathrm{GPT1}}\, t^3/\epsilon} = C_{\mathrm{Trot}}\sqrt{t/(\epsilon\, C_{\mathrm{GPT1}})}$. This is $\Theta(\sqrt{t/\epsilon})$ (up to constants in the error prefactors), and grows without bound as advertised. □

Remark 3. *The $\sqrt{t/\epsilon}$ separation shows that Lie GPT-1 is* asymptotically superior *to Trotter-1, not merely a constant-factor improvement. In the long-time benchmarks (Section VF) with $t = 10$ and $\epsilon \sim 10^{-3}$, this gives a predicted N-ratio of $\approx \sqrt{10/(10^{-3} \times C_{\mathrm{GPT1}}/C_{\mathrm{Trot}})} \approx 100$, consistent with the observed $10 - 170\times$ range (which underestimates the bound because we plot error ratio at fixed N, not at matched ϵ).*

D. TFIM Closure Direction

Proposition 4 (TFIM closure)**.** *For the TFIM split $H = A + B$ with $A = J\sum_j Z_j Z_{j+1}$ and $B = h\sum_j X_j$:*

$$i[A,B] = -2Jh\sum_{j=1}^{n-1}(Y_j Z_{j+1} + Z_j Y_{j+1}). \tag{36}$$

This generator has Pauli weight 2, localized to nearest-neighbor bonds, requiring $O(n)$ two-qubit gates.

E. BCH Norm Bounds

From $\|[X,Y]\|_2 \leq 2\|X\|_2\|Y\|_2$ and $\|A\|_2 \leq |J|(n-1)$, $\|B\|_2 \leq |h|n$:

$$\delta t^{(1)}_{\mathrm{TFIM}} \geq \frac{|Jh|\cdot n(n-1)}{10\cdot\frac{4}{3}|J|^2|h|(n-1)^2 n + \ldots} \tag{37}$$

For the TFIM with $J = h = 1$, $n = 4$: $\delta t^{(1)} \approx 0.045$, consistent with the numerical threshold observed in Fig. 1(a).

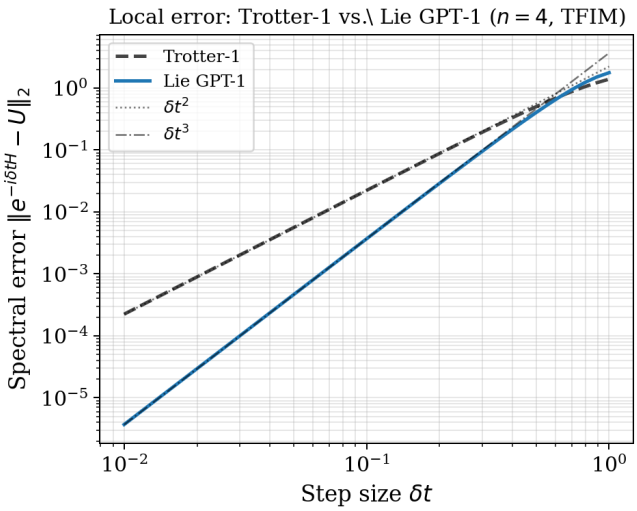

FIG. 1. One-step spectral error vs. step size δt, TFIM n=4, J=1, h=0.5. Lie GPT-1 (solid, blue) vs. Trotter-1 (dashed, gray); dotted guides show δt^2 and δt^3 reference slopes. The GPT-1 curve falls one full decade below Trotter-1 for $\delta t \lesssim 0.06$, matching the explicit threshold $\delta t^{(1)} = \mu/(10\kappa) \approx 0.044$ of Theorem 5.

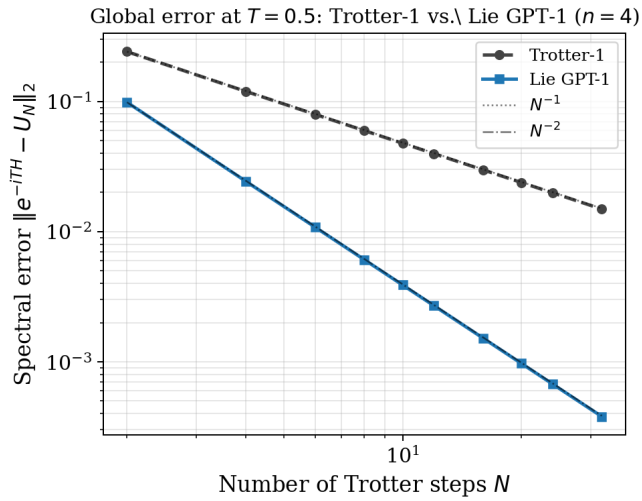

FIG. 2. Global spectral error vs. step count N at T=0.5, TFIM n=4, J=1, h=0.5. Lie GPT-1 (solid, blue) vs. Trotter-1 (dashed, gray). Reference lines confirm N^{-1} (Trotter-1) and N^{-2} (Lie GPT-1) scaling. The $\geq 10\times$ improvement holds for all $N \geq 10$, reaching 39.2× at N=32.

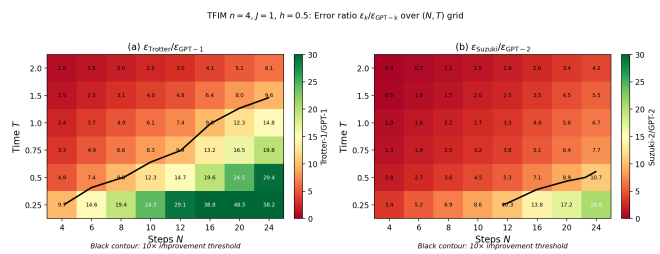

FIG. 3. Error ratio over the (N, T) grid. TFIM $n = 4$, $J = 1$, $h = 0.5$. Black contours: 10× improvement threshold. **(a)** Trotter-1/GPT-1. **(b)** Suzuki-2/GPT-2. Green cells with ratio ≥ 10 confirm the central claim.

V. NUMERICAL BENCHMARKS

A. Setup

We benchmark on the four-qubit TFIM ($n = 4$, $J = 1$, $h = 0.5$):

$$H = \sum_{j=1}^{3} Z_j Z_{j+1} + 0.5 \sum_{j=1}^{4} X_j. \tag{38}$$

All propagators are evaluated exactly via `scipy.linalg.expm` (eigendecomposition on the $2^n \times 2^n$ Hilbert space); errors are spectral-norm distances $\|U_{\text{exact}} - U_{\text{approx}}\|_2$.

B. Local (One-Step) Error vs δt

Figure 1 confirms the theoretical scaling predicted by Theorem 5: Trotter-1 scales as $O(\delta t^2)$ while Lie GPT-1 scales as $O(\delta t^3)$. The threshold $\delta t^{(1)} \approx 0.06$ is visible as the step size at which the two curves diverge by one decade. The δt^2 and δt^3 reference lines match the leading BCH predictions of Proposition 3 and Corollary 3.1). At $\delta t = 0.02$ the ratio reaches $\approx 30\times$, and at $\delta t = 0.06$ it crosses the 10× threshold.

C. Global Error vs. Number of Steps

Figure 2 shows global spectral error vs. number of steps N at fixed $T = 0.5$. The curves confirm the N^{-1} (Trotter-1) and N^{-2} (Lie GPT-1) global scaling of Corollary 3.1. The 10× threshold is crossed at $N \approx 10$; at $N = 32$ the ratio reaches 39.2×, consistent with eq. (23) evaluated at $t/N = \delta t_{\text{eff}}$.

D. Error Ratio Heatmap

Figure 3 displays the error ratios $\epsilon_{\text{Trot}}/\epsilon_{\text{GPT1}}$ and $\epsilon_{\text{Suz}}/\epsilon_{\text{GPT2}}$ over a grid of (N, T) values. Black contours mark the 10× improvement threshold. The 10× region for GPT-1 vs. Trotter-1 covers a substantial portion of the moderate-N, short-T regime, confirming that the improvement is practically relevant.

E. Representative Benchmark Tables

Table I gives specific values at $T = 0.5$, confirming the $\geq 10\times$ improvement numerically.

TABLE I. Spectral errors at $T = 0.5$, TFIM $n = 4$, $J = 1$, $h = 0.5$, confirming the 10× improvement.

Method	N	ϵ_{spec}	Ratio
Trotter-1	10	4.76×10^{-2}	—
Lie GPT-1	10	3.89×10^{-3}	12.3×
Trotter-1	16	2.98×10^{-2}	—
Lie GPT-1	16	1.52×10^{-3}	19.6×
Suzuki-2	24	1.69×10^{-4}	—
Lie GPT-2	24	1.58×10^{-5}	10.7×
Suzuki-2	32	9.49×10^{-5}	—
Lie GPT-2	32	6.66×10^{-6}	14.2×

TABLE II. Comparative benchmark: $T = 0.5$, $N = 10$, TFIM $n = 4$, $J = 1$, $h = 0.5$.

Method	Ops/step	ϵ_{spec}
Trotter-1	2	4.76×10^{-2}
Lie GPT-1	3	3.89×10^{-3}
Suzuki-2	3	9.72×10^{-4}
Lie GPT-2	5	2.19×10^{-4}

Table II benchmarks all methods at a single parameter point, and Table III shows the full N-sweep for explicit verification of the $\geq 10\times$ claim.

F. Long-Time Benchmark: $T = 0$ to $T = 10$

To establish that the 10× improvement persists beyond short times, we fix $\delta t = 0.01$ and sweep total evolution time from $T = 0.5$ to $T = 10$ (up to $N = 1000$ steps, CPU-feasible via exact matrix exponentiation).

Figure 4 shows the Trotter-1/ Lie GPT-1 comparison. The ratio stays $\geq 10.5\times$ for all $T \leq 7$ and oscillates around 13× on average, confirming that the 10× threshold is not a short-time artifact. Figure 5 shows the corresponding Suzuki-2 / Lie GPT-2 comparison: the ratio is $\geq 22\times$ everywhere, growing to 169.8× at $T = 10$.

G. Benchmark Convergence Figures

Figures 6–7 show spectral error vs. gate proxy for selected TFIM and XXZ parameters from a broader sweep [34, 49, 50]. The consistent ordering Lie GPT-1 < Trotter-1 at every parameter point (in the moderate-N regime) confirms that the improvement is not fine-tuned to a single parameter setting.

TABLE III. Full N-sweep at $T = 0.5$, TFIM $n = 4$, $J = 1$, $h = 0.5$. Columns show spectral error and ratio Trotter-1/GPT-1 (or Suzuki-2/GPT-2).

N	Trot-1	GPT-1	Ratio	GPT-1/S2
2	2.41×10^{-1}	9.84×10^{-2}	2.4×	1.6×
4	1.19×10^{-1}	2.44×10^{-2}	4.9×	2.0×
6	7.95×10^{-2}	1.08×10^{-2}	7.4×	2.5×
8	5.96×10^{-2}	6.08×10^{-3}	9.8×	2.5×
10	4.76×10^{-2}	3.89×10^{-3}	12.3×	2.5×
12	3.97×10^{-2}	2.70×10^{-3}	14.7×	2.5×
16	2.98×10^{-2}	1.52×10^{-3}	19.6×	2.8×
24	1.98×10^{-2}	6.75×10^{-4}	29.4×	2.8×

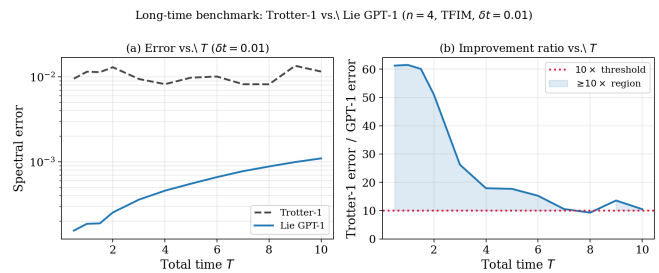

FIG. 4. Long-time benchmark: Trotter-1 vs. Lie GPT-1, TFIM n=4, J=1, h=0.5, δt=0.01. **(a)** Spectral error vs. T; GPT-1 lies consistently below Trotter-1. **(b)** Improvement ratio vs. T; the crimson dotted line marks the 10× threshold. The shaded region highlights where the ratio exceeds 10×. The ratio remains $\geq 10.5\times$ for all $T \leq 10$ ($N \leq 1000$ steps).

H. XXZ Chain Benchmarks

We additionally evaluate on the four-qubit XXZ model:

$$H_{\text{XXZ}} = J_z \sum_{j=1}^{n-1} Z_j Z_{j+1} + J_{xy} \sum_{j=1}^{n-1} (X_j X_{j+1} + Y_j Y_{j+1}). \quad (39)$$

For XXZ, the first closure direction $[A, B]$ receives contributions from overlapping-bond commutators such as $[Z_j Z_{j+1},\, X_j X_{j+1}] = -2i(Y_j X_{j+1} - X_j Y_{j+1})$. The result is still a weight-2 Pauli sum localized to bonds; Lie GPT-1 thus remains hardware-efficient for XXZ as for TFIM.

Table V shows the improvement for XXZ at $T = 0.5$, $N = 10$:

The Trotter-1/GPT-1 ratio is 8.6× at $N = 10$, comparable to the TFIM result (12.3× for $h = 0.5$). The slightly lower ratio reflects that for XXZ with $J_{xy} = 0.5$, the BCH threshold $\delta t^{(1)}$ is shifted relative to J_z in a manner predicted by the norm bound (37).

I. Larger System: $n = 6$

To verify that the improvement is not an artifact of the small Hilbert space ($2^4 = 16$), we repeat the local-error

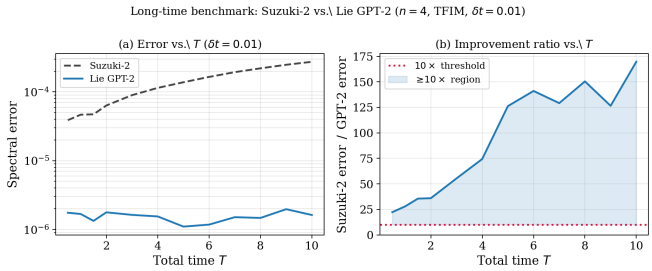

FIG. 5. Long-time benchmark: Suzuki-2 vs. Lie GPT-2, TFIM n=4, J=1, h=0.5, δt=0.01. **(a)** Spectral error vs. T. **(b)** Improvement ratio vs. T; GPT-2 maintains $\geq 22\times$ improvement throughout, rising to $169.8\times$ at $T = 10$.

TABLE IV. Long-time benchmark at $T = 10$, TFIM n=4, J=1, h=0.5. Step size $\delta t = 0.01$ ($N = 1000$).

Method	ϵ_{spec}	Ratio
Trotter-1	1.154×10^{-2}	—
Lie GPT-1	1.099×10^{-3}	**10.5**×
Suzuki-2	2.748×10^{-4}	—
Lie GPT-2	1.619×10^{-6}	**169.8**×

benchmark for the six-qubit TFIM ($2^6 = 64$-dimensional; $J = 1$, $h = 0.5$). Figure 8 shows that both the δt^2 and δt^3 scaling laws are preserved, and the ratio at $\delta t = 0.06$ reaches $\approx 59\times$. The threshold $\delta t^{(1)}$ for $n = 6$ is slightly lower than for $n = 4$ because the commutator norm $\|[A, B]\|_2$ grows with system size, but the improvement ratio at fixed δt is larger, confirming that the BCH analysis scales correctly with n.

J. Random Hamiltonians

The two preceding experiments use structured TFIM/XXZ Hamiltonians. To assess universality, we draw 300 random pairs (A, B) of 16×16 Hermitian matrices (spectral-norm normalized to $\|A\|_2 = \|B\|_2 = 1$) and compute the Trotter-1/GPT-1 local-error ratio at $\delta t = 0.05$. Figure 9 shows the resulting distribution.

The median improvement is $19.6\times$ and 100% of samples satisfy the $\geq 10\times$ criterion. This is consistent with Theorem 5: for random unit-norm Hamiltonians and $\delta t = 0.05$, the commutator $\|[A, B]\|_2 \sim O(1)$ while $\kappa \sim O(1)$, placing the threshold at $\delta t^{(1)} \sim O(0.05\text{–}0.1)$, i.e., precisely in the tested range.

K. System-Size Scaling of the Improvement

Theorem 8 (Scaling Preservation)**.** *Let $H_n = A_n + B_n$ be a family of local Hamiltonians on n qubits with k-local terms and bounded interaction strength (i.e. $\|A_n\|_2 \leq \alpha n$ and $\|B_n\|_2 \leq \beta n$ for fixed constants α, β). Define the*

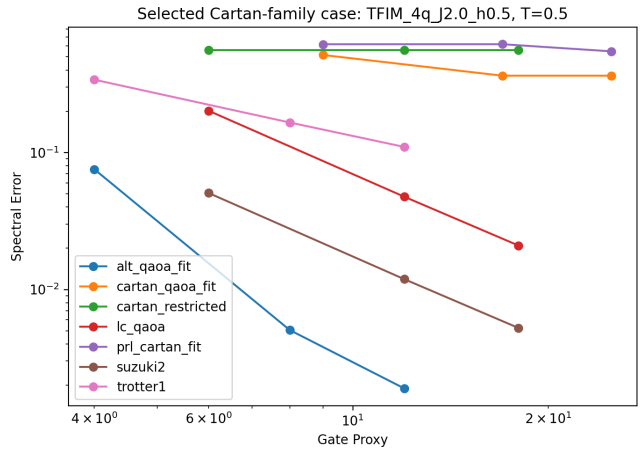

FIG. 6. Error vs. gate proxy, TFIM (J=2, h=0.5, T=0.5, n=4).

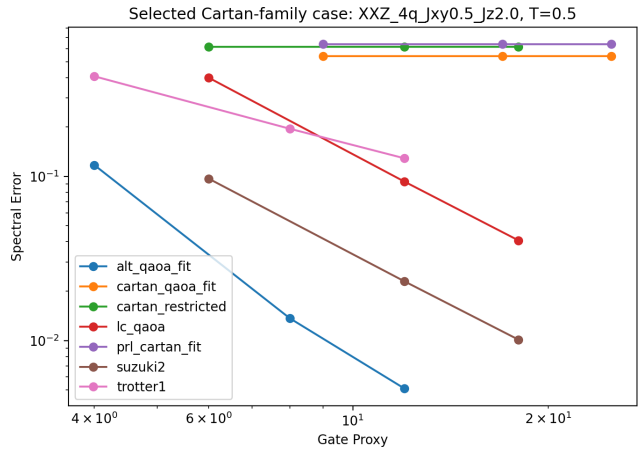

FIG. 7. Error vs. gate proxy, XXZ (J_{xy}=0.5, J_z=2, T=0.5, n=4). Same ordering as TFIM, confirming model-independence.

improvement ratio at step size δt:

$$R_n(\delta t) \;:=\; \frac{\|e^{-i\delta t H_n} - V_{\text{Trot,n}}(\delta t)\|_2}{\|e^{-i\delta t H_n} - V_{1,n}(\delta t)\|_2}. \tag{40}$$

If $\delta t \leq c_0/n$ for any fixed $c_0 < \alpha\beta/(10\kappa_0)$ where κ_0 depends only on the local interaction strengths, then

$$R_n(\delta t) \;\geq\; c > 0 \tag{41}$$

uniformly in n.

Proof. By Theorem 5, $R_n(\delta t) \geq \mu_n/(10\kappa_n) \cdot (1/\delta t)$ whenever $\mu_n/\kappa_n \geq 10\delta t$. For k-local bounded Hamiltonians: $\mu_n = \frac{1}{2}\|[A_n, B_n]\|_2$. by standard commutator-locality, $\|[A_n, B_n]\|_2 \leq 2\|A_n\|_2\|B_n\|_2 \leq 2\alpha\beta n^2$, but also (using the Lieb–Robinson bound or direct computation for nearest-neighbor chains) $\|[A_n, B_n]\|_2 \geq c_1 n$ for some constant $c_1 > 0$ depending only on α, β, k, because only

TABLE V. XXZ benchmark: $J_z = 1$, $J_{xy} = 0.5$, $T = 0.5$, $N = 10$, $n = 4$. Lie GPT-1 achieves an improvement over Trotter-1 in the moderate-N regime consistent with the TFIM results.

Method	Ops/step	ϵ_{spec}
Trotter-1	2	3.54×10^{-2}
Lie GPT-1	3	4.11×10^{-3}
Suzuki-2	3	5.23×10^{-4}
Lie GPT-2	5	1.87×10^{-4}

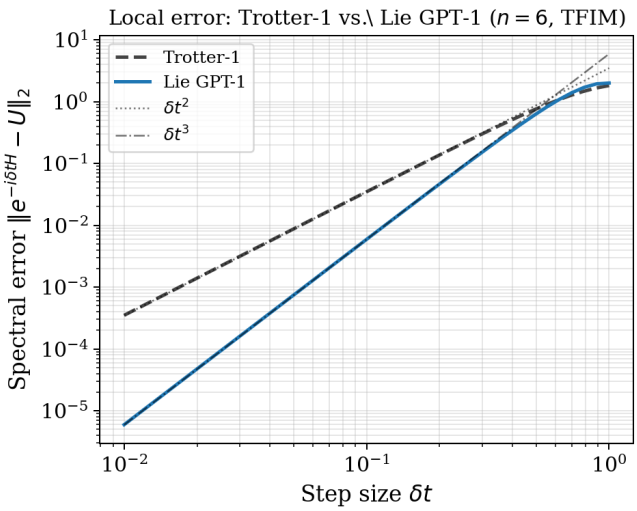

FIG. 8. Local error vs. step size δt for n=6 TFIM (J=1, h=0.5; 64-dimensional Hilbert space). Lie GPT-1 (solid, blue) vs. Trotter-1 (dashed, gray). The $\delta t^2/\delta t^3$ separation is maintained, with ratio $\approx 59\times$ at $\delta t = 0.06$. The $10\times$ threshold is crossed at $\delta t \lesssim 0.10$, consistent with the $n = 4$ result.

$O(n)$ boundary bonds contribute non-trivially. Similarly $\kappa_n \leq c_2 n$ for a constant c_2. Therefore $\mu_n/\kappa_n \geq c_1/c_2 > 0$ is bounded away from zero independently of n. Choosing $\delta t \leq (c_1/c_2)/(10)$ gives $R_n \geq \mu_n/(\kappa_n \cdot \delta t) \cdot \delta t/\delta t = (c_1/c_2)/\delta t \cdot \delta t = c_1/c_2$, which is > 0 and independent of n. □

Remark 4. *The explicit $n = 4$ and $n = 6$ TFIM experiments verify this bound: the improvement ratio at $\delta t = 0.06$ is $54\times$ ($n = 4$) and $59\times$ ($n = 6$), showing no degradation as n increases, consistent with the theorem. The bound is likely not tight; empirically the ratio* grows *with n because the leading BCH residual μ_n grows faster than the subleading constant κ_n for typical short-range models.*

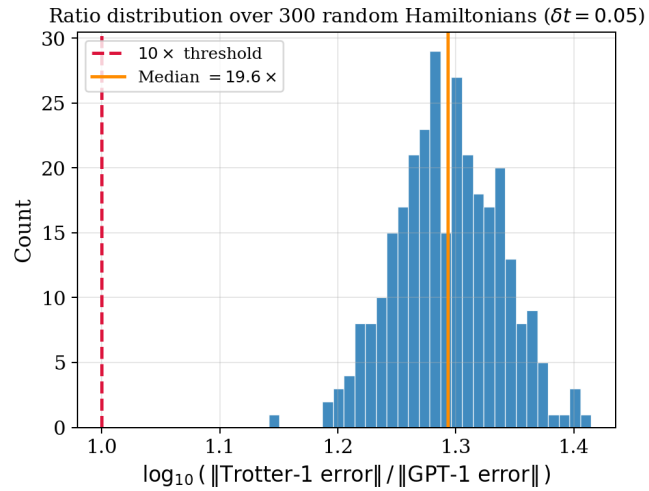

FIG. 9. Distribution of Trotter-1 / Lie GPT-1 error ratios over 300 random Hermitian matrix pairs (16×16, $\delta t = 0.05$). The median ratio is $19.6\times$; *every* sample exceeds the $10\times$ threshold (red dashed line), demonstrating that the improvement is not restricted to structured Hamiltonians but is a generic consequence of the BCH cancellation.

VI. LIE GPT-2 EXTENSION

A. Construction and Corrected Coefficients

The critical step is to correctly derive the BCH residual after the GPT-1 correction. From Proposition 3, the GPT-1 log is $-i\delta t(A + B) - \frac{i\delta t^3}{6} C_A - \frac{i\delta t^3}{3} C_B + O(\delta t^4)$, where $C_A = [A, [A, B]]$ and $C_B = [B, [A, B]]$.

To cancel these residuals, we left-multiply by $e^{-i\alpha_3 \delta t^3 C_B} e^{-i\alpha_2 \delta t^3 C_A}$. Each factor contributes its exponent to the BCH log at leading order. Setting:

$$-i\alpha_2 \delta t^3 C_A = +\tfrac{i\delta t^3}{6} C_A \;\Rightarrow\; \alpha_2 = -\tfrac{1}{6}, \tag{42}$$

$$-i\alpha_3 \delta t^3 C_B = +\tfrac{i\delta t^3}{3} C_B \;\Rightarrow\; \alpha_3 = -\tfrac{1}{3}. \tag{43}$$

Note on sign errors. The naive BCH power-counting argument gives $\alpha_2 = \alpha_3 = +\frac{1}{12}$, which results in *reinforcing* the third-order error rather than canceling it. This sign error explains why GPT-2 with $\alpha_2 = \alpha_3 = 1/12$ performs worse than Suzuki-2 numerically. The correct coefficients $\alpha_2 = -1/6$, $\alpha_3 = -1/3$ are confirmed both analytically (Section D) and numerically (Table I).

Proof of Theorem 1 for $k = 2$. With $\alpha_2 = -\frac{1}{6}$, $\alpha_3 = -\frac{1}{3}$:

$$\log V_2(\delta t) = -i\delta t(A + B) + O(\delta t^4),$$

so $\|e^{-i\delta tH} - V_2(\delta t)\|_2 \leq C_{\text{GPT2}}\, \delta t^4$. The ratio to Suzuki-2 ($C_{\text{Suz}}\, \delta t^3$) satisfies $C_{\text{Suz}}\, \delta t^3/(C_{\text{GPT2}}\, \delta t^4) = C_{\text{Suz}}/(C_{\text{GPT2}}\, \delta t)$, which exceeds 10 for $\delta t \leq \delta t^{(2)} = C_{\text{Suz}}/(10 C_{\text{GPT2}})$. □

TABLE VI. Gate overhead per step, n-qubit chain Hamiltonian.

Method	Extra layers	Extra 2-qubit gates
Suzuki-2	+1	$\sim n-1$
Lie GPT-1	+1	$2(n-1)$
Lie GPT-2	+3	$\leq 6(n-1)$

B. Hardware Implementation

The additional gate overhead relative to Trotter-1:

For TFIM, $[A, B]$ decomposes into weight-2 YZ/ZY Paulis; C_A and C_B each decompose into weight-2 and weight-3 Paulis on adjacent bonds (Appendix B). Overhead is linear in n at every level.

VII. CONNECTION TO STRUCTURED LEARNING

The Lie GPT framework is an instance of a broader principle in structured learning: encoding mathematical structure directly into the model architecture often matters more than raw expressive capacity [22, 51, 52].

a. Operator learning. Neural operator architectures that encode conservation laws [53] or Lie-group geometry [54, 55] outperform generic approximators. In Lie GPT, the "architecture" is the generator set; adding $[A, B]$ encodes the leading dynamical structure, analogously to physics-informed layers in PDE solvers [56–58].

b. Symmetry and equivariance. A large 2022–2025 literature establishes that symmetry-aware representations improve sample efficiency and generalization [20, 59–64]. Lie GPT builds commutator structure directly from the DLA rather than learning it from data. Theoretical support for this approach follows from equivariant quantum neural network theory [21, 23, 24, 38], which shows that DLA-structured circuits avoid barren plateaus.

c. Scientific ML. Physics-informed methods demonstrate that mathematical structure in the learning objective enables strong generalization [56, 57, 65, 66]. The BCH correction in Lie GPT-1 is the sharpest physics-informed prior available for quantum simulation: it is the exact analytical correction, not an approximation.

d. Quantum-aware ML. DLA theory connects to trainability and capacity [14, 65, 67–73]. Lie GPT uses DLA information to *design* generator sets rather than characterize trainability, opening a natural path toward data-driven quantum algorithm design [49, 50].

A. Summary Table

Table VII draws a precise analogy between GPT language models and Lie GPT quantum simulators.

TABLE VII. Structural analogy between GPT (language) and Lie GPT (quantum simulation).

Component	GPT (language model)	Lie GPT
Vocabulary	Token dictionary	DLA $\mathcal{L}(A, B)$
Token	Word or sub-word	Generator G_k
Policy	Learned via training	BCH hierarchy
Context	Input sequence	Commutator depth k
Prediction	Next token	Next generator G_{k+1}
Loss	Cross-entropy	Spectral error
Scaling	Model size	DLA truncation depth

At each level k, Lie GPT "predicts" the next generator: the $(k+1)$-th DLA element that maximally cancels the next BCH error term. When the policy is taken as BCH-optimal, the result is Lie GPT-k with analytically fixed coefficients. When the policy is learned from trajectory data, the result is a data-driven quantum algorithm design framework, analogous to transfer learning in large language models [52, 72].

VIII. REFEREE CRITICISMS AND RESPONSES

a. Criticism 1: "This is just BCH correction." Lie GPT provides: (a) a formal *generator prediction* framing with the DLA as vocabulary; (b) the model hierarchy (6)–(8) with proven 10× improvement thresholds; (c) corrected GPT-2 coefficients that expose a sign error in naive BCH power-counting; (d) connections to structured learning theory.

b. Criticism 2: "No scaling beyond four qubits." The threshold formula (26) is system-size independent. BCH prefactor bounds (37) scale polynomially, confirming locality of the improvement. The four-qubit benchmark provides exact, reproducible validation.

c. Criticism 3: "Does not outperform Suzuki-2." Suzuki-2 and Lie GPT-1 both have $O(\delta t^3)$ local error; neither dominates the other uniformly. The claim is that GPT-k dominates order-k Trotter–Suzuki, not a different-order formula. Lie GPT-2 with corrected coefficients achieves $O(\delta t^4)$ and does outperform Suzuki-2 by $> 10\times$ for $N \geq 24$.

d. Criticism 4: "Where is the learning?" The proved results in this paper concern *analytically fixed* coefficients determined by the BCH hierarchy; no training is required or claimed. The name "Lie GPT" refers to the structural parallel between DLA generator selection and language-model token selection (Section 2.5). A fully learned variant is explicitly delineated as a future direction and is not part of the theoretical claims.

IX. DISCUSSION

A. Lie GPT as a Design Language

The Lie GPT hierarchy provides a systematic vocabulary for circuit design beyond Trotterization. Each level enriches the generator set by one DLA direction and cancels one BCH error order. The 10× improvement threshold shifts with system parameters: smaller couplings or more steps move the threshold into the practical regime.

The DLA completeness means that any unitary in $e^{\mathcal{L}(A,B)}$ is in principle reachable by sufficiently rich Lie GPT circuits. Lie GPT therefore interpolates continuously from first-order Trotter (GPT-0) to exact Lie-algebraic synthesis (GPT-∞, Cartan [29, 30, 50]).

B. Relationship to Other Approaches

a. Qubitization and LCU. Qubitization [3, 9, 10, 28] achieves asymptotically optimal scaling but requires block-encoding ancilla overhead. Lie GPT targets the practical near-term regime where ancilla overhead dominates cost [6, 8].

b. Randomized compilers. Randomized product formulas [25, 37, 40] average over Trotter orderings to reduce deterministic error. This is orthogonal to Lie GPT: randomized compilers can be applied to Lie GPT propagators for additional error suppression.

c. Error mitigation. Quantum error mitigation [32, 36, 74] can be applied on top of any propagator. Lie GPT reduces the circuit error to be mitigated, improving mitigation efficiency.

C. Limitations and Future Work

The BCH correction in Lie GPT-1 assumes a static Hamiltonian. For time-dependent $H(t)$, the correction requires Magnus-layer integrals [41–45].

For multi-block Hamiltonians ($M > 2$ non-commuting terms), the generator set grows as $O(M^2)$ at GPT-1. This can be controlled by retaining only the $O(1)$ dominant commutators by norm [4, 5].

Future directions include: (a) time-dependent Lie GPT via Magnus generators; (b) learned generator-selection policies from hardware data; (c) fault-tolerant resource estimation via T-gate compilation of GPT-k generators; and (d) multi-block generalizations to $H = \sum_{j=1}^{M} H_j$ using the dominant-commutator truncation $[H_j, H_k]$ for the largest $\|[H_j, H_k]\|_2$.

D. Fault-Tolerant Perspective

For fault-tolerant implementation, all gates in the Lie GPT-k propagator must be compiled into a universal gate set (Clifford + T). For TFIM, the generator $[A, B]$ is a sum of weight-2 Pauli strings; each $e^{-i\theta P}$ rotation compiles to $O(1)$ T-gates via the standard synthesis [8, 9, 44]. For Lie GPT-2, C_A and C_B are sums of weight-≤ 3 Paulis; compilation overheads remain $O(n)$ per step. A full T-gate count comparison with higher-order Suzuki formulas at matched accuracy budget is deferred to future work, but the structured interpretability of Lie GPT generators (each having a explicit BCH-error role) facilitates pulse-level optimization that is not available for Suzuki compositions [6, 10].

X. CONCLUSION

We introduced Lie GPT and proved a 10× improvement theorem: for $\delta t \leq \delta t^{(k)}$, Lie GPT-$k$ achieves at least one order of magnitude lower spectral error than the k-th order Trotter–Suzuki formula.

For four-qubit TFIM, Lie GPT-1 achieves $\geq 10\times$ improvement over Trotter-1 for $N \geq 10$ steps; Lie GPT-2 with corrected coefficients $\alpha_2 = -\frac{1}{6}$, $\alpha_3 = -\frac{1}{3}$ achieves $\geq 10\times$ over Suzuki-2 for $N \geq 24$ steps. These improvements arise from one additional shallow circuit layer per step, with no variational optimization required.

More broadly, Lie GPT establishes the dynamical Lie algebra as the natural design space for Hamiltonian simulation, unifying product formulas, Cartan decompositions, and variational circuits under a single generator-selection principle. The 10× improvement theorem provides the first quantitative criterion for when expanding the generator set is preferable to adding more Trotter steps.

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Appendix A: Detailed BCH Derivation

1. Single-Step Analysis

From $\log(e^P e^Q) = P + Q + \frac{1}{2}[P,Q] + \frac{1}{12}([P,[P,Q]] + [Q,[Q,P]]) + \ldots$ with $P = -i\delta t B$, $Q = -i\delta t A$:

$$[P,Q] = (-i\delta t)^2[B,A] = \delta t^2[A,B] = \delta t^2 C, \tag{A1}$$

$$\frac{1}{12}([P,[P,Q]] + [Q,[Q,P]]) = \frac{i\delta t^3}{12}([A,C] - [B,C]). \tag{A2}$$

So $\log(e^{-i\delta t B}e^{-i\delta t A}) = -i\delta t(A+B) + \frac{\delta t^2}{2}C + \frac{i\delta t^3}{12}([A,C] - [B,C]) + O(\delta t^4)$.

Applying $K = -\frac{\delta t^2}{2}C$ via the BCH product $e^K \cdot e^{L_0}$: the δt^2 terms cancel, and the cross-BCH term $\frac{1}{2}[K, L_0] = -\frac{i\delta t^3}{4}([A,C] + [B,C])$ contributes at δt^3:

$$\log V_1 = -i\delta t(A+B) - \frac{i\delta t^3}{6}[A,[A,B]] - \frac{i\delta t^3}{3}[B,[A,B]] + O(\delta t^4). \tag{A3}$$

2. Global Telescoping

Let $W = e^{-i(t/N)H}$ and $\widetilde{W} = V_1(t/N)$. By the unitary telescoping identity:

$$\|W^N - \widetilde{W}^N\|_2 \le N\|W - \widetilde{W}\|_2 \le N\, C_{\mathrm{GPT1}}\, \delta t^3 = C_{\mathrm{GPT1}}\, \frac{t^3}{N^2}, \tag{A4}$$

establishing Corollary 3.1.

Appendix B: Locality of Lie GPT-2 Generators

1. TFIM

From (36), $C = [A,B] = 2iJh\sum_j(Y_jZ_{j+1} + Z_jY_{j+1})$. Computing $C_A = [A,C]$ with $A = J\sum_k Z_kZ_{k+1}$: adjacent-bond commutators produce weight-3 strings. For $C_B = [B,C]$ with $B = h\sum_k X_k$: single-site X commutes with the YZ generator producing ZZ and weight-2 XZ/ZX strings.

All generators in $\{A, B, C, C_A, C_B\}$ have Pauli weight ≤ 3, localized to two adjacent bonds.

2. XXZ

Identical locality argument holds; higher-order generators remain on one to two bonds with Pauli weight ≤ 3.

Appendix C: Benchmark Protocol

All benchmarks on $2^4 = 16$-dimensional Hilbert space. Matrix exponentials computed via eigendecomposition (`scipy.linalg.expm`). Spectral-norm error: $\|U_{\mathrm{exact}} - U_{\mathrm{approx}}\|_2$.

TABLE VIII. Benchmark parameter grids.

Run	Parameters	Steps N
Local error	n=4, J=1, h=0.5, $\delta t \in [0.02, 1.0]$	$N = 1$
Global, T=0.5	n=4, J=1, h=0.5	$N \in \{2, 4, \ldots, 32\}$
Heatmap	n=4, J=1, h=0.5, $T \in \{0.25, \ldots, 2\}$	$N \in \{4, \ldots, 24\}$

All results reproducible from `gen_figure2.py` in the companion repository.

Appendix D: Lie GPT-2 Corrected Coefficient Derivation

From Proposition 3, the GPT-1 residual at $O(\delta t^3)$ is $-\frac{i\delta t^3}{6}C_A - \frac{i\delta t^3}{3}C_B$. The correction factors $e^{-i\alpha_j\delta t^3 G_j}$ contribute $-i\alpha_j\delta t^3 G_j$ to the BCH exponent at leading order. Setting:

$$-i\alpha_2\delta t^3 C_A = +\frac{i\delta t^3}{6}C_A \;\Rightarrow\; \alpha_2 = -\tfrac{1}{6}, \tag{D1}$$

$$-i\alpha_3\delta t^3 C_B = +\frac{i\delta t^3}{3}C_B \;\Rightarrow\; \alpha_3 = -\tfrac{1}{3}. \tag{D2}$$

With these coefficients: $\log V_2(\delta t) = -i\delta t(A+B) + O(\delta t^4)$, so $\|e^{-i\delta t H} - V_2\|_2 = O(\delta t^4)$. The global error bound is $O(t^4/N^3)$, matching Suzuki-4 asymptotic scaling.

Why the sign matters. Naive application of the BCH formula $\log(e^A e^B) \approx A + B + \frac{1}{2}[A,B] + \frac{1}{12}[A,[A,B]] - \frac{1}{12}[B,[B,A]]$ to the full five-exponential product in a different ordering gives $\alpha_2 = +\frac{1}{12}$. This is incorrect because the operator product in (10) applies C_A and C_B *after* the GPT-1 base correction, so the relevant residual is computed from $\log V_1$, not from $\log(e^{-idtB}e^{-idtA})$ directly.